Wenyuan (Harry) Huang

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Education

University of Wisconsin Madison — *Jan 2024 – May 2027 (Expected)*
BS with Honor degree in Computer Science, Mathematics

- GPA: 3.87/4.0
- **Coursework:** Advanced Topics in Reinforcement Learning, Nonlinear Optimization, Deep Learning for Computer Vision

Experience

Undergraduate Research Assistant — *Madison, WI*
University of Wisconsin Madison - Computer Science Department — *Jan 2025 – Present*

- Conducting research in Reinforcement Learning (RL) with Prof. Josiah Hanna.

Projects

Multi-task RL with Distributionally Robust Optimization — *Feb 2025 - Present*
MultiTaskRL

- Applied Distributionally Robust Optimization to trajectory collections for various RL algorithms, including PPO, DDPG.
- Implemented improved data collection and task sampling methods based on MTBench and CleanRL. Outperform traditional multitask implementation for about 20% on Gridworld and Bandit, also greater success rate in PointMaze.
- Used MTRL techniques to reduce negative transfer between tasks, such as multi-head policy and one-hot task embedding.
- Planning to submit as a proceedings paper on ICML.

RoboCup Humanoid Soccer Project — *Oct 2025 - Present*

- Will develope an RL policy to enable the Booster T1 robot to perform precise and robust kicking actions. Train the policy to handle noisy ball localization and dynamic humanoid balance.
- Planning to build upon the Wistex United RoboCup codebase and formalized the environment as an MDP for reproducible benchmarking.
- Evaluate accuracy, goal-scoring rate, and robustness under Gaussian perception noise.
- Aiming to demonstrate RL-based motion control outperforming rule-based approaches in RoboCup scenarios.

Honor & Awards

Champion of ICPC North Central NA Regional Contest — Nov 2024
Ranked **1st among 96 ICPC teams** in the North Central North America region; the only team to solve 9 problems.

Nicholas D. Yankaitis Memorial Scholarship — May 2025
One of the most prestigious awards recognizing academic excellence among undergraduate Computer Sciences majors.

Technologies

Languages: Python, Java, C++, Markdown, LaTeX, HTML, CSS, JavaScript

Technology skills: Reinforcement Learning, Multi-task Learning, Deep Learning, Computer Vision, Algorithms, Data Structures